

# Target and stitched multiple-shooting estimate

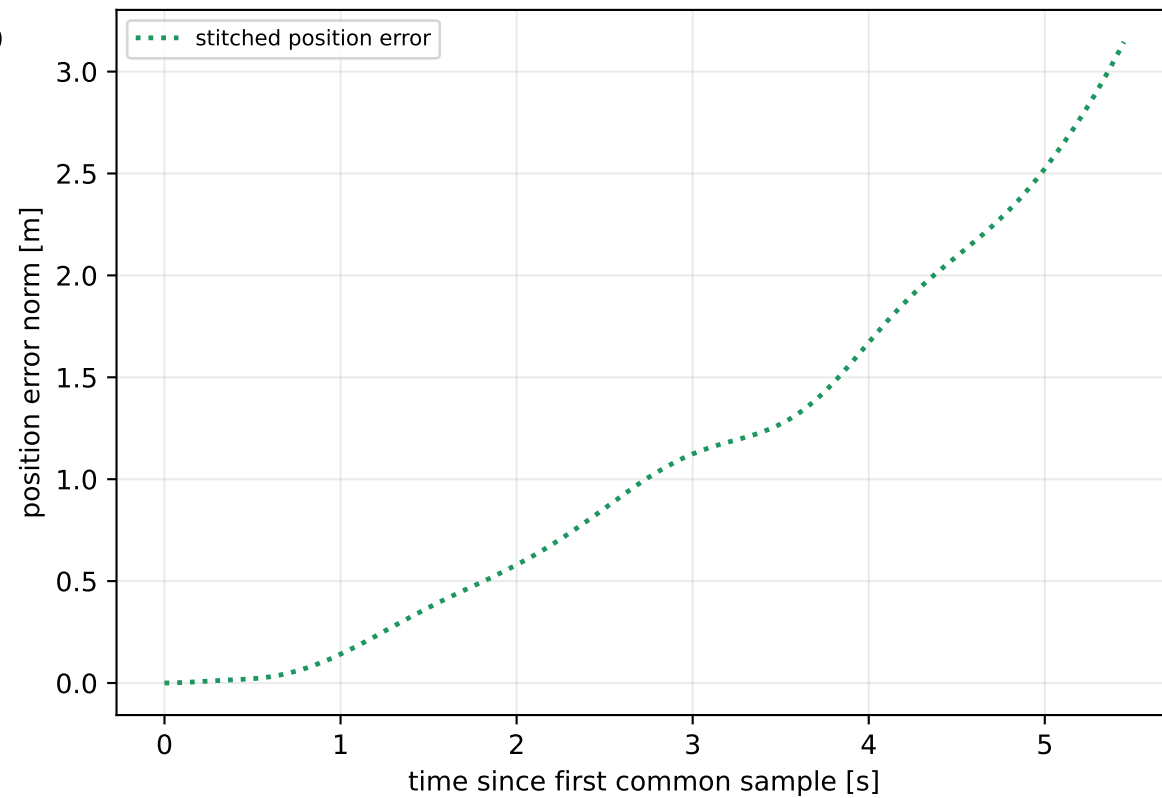
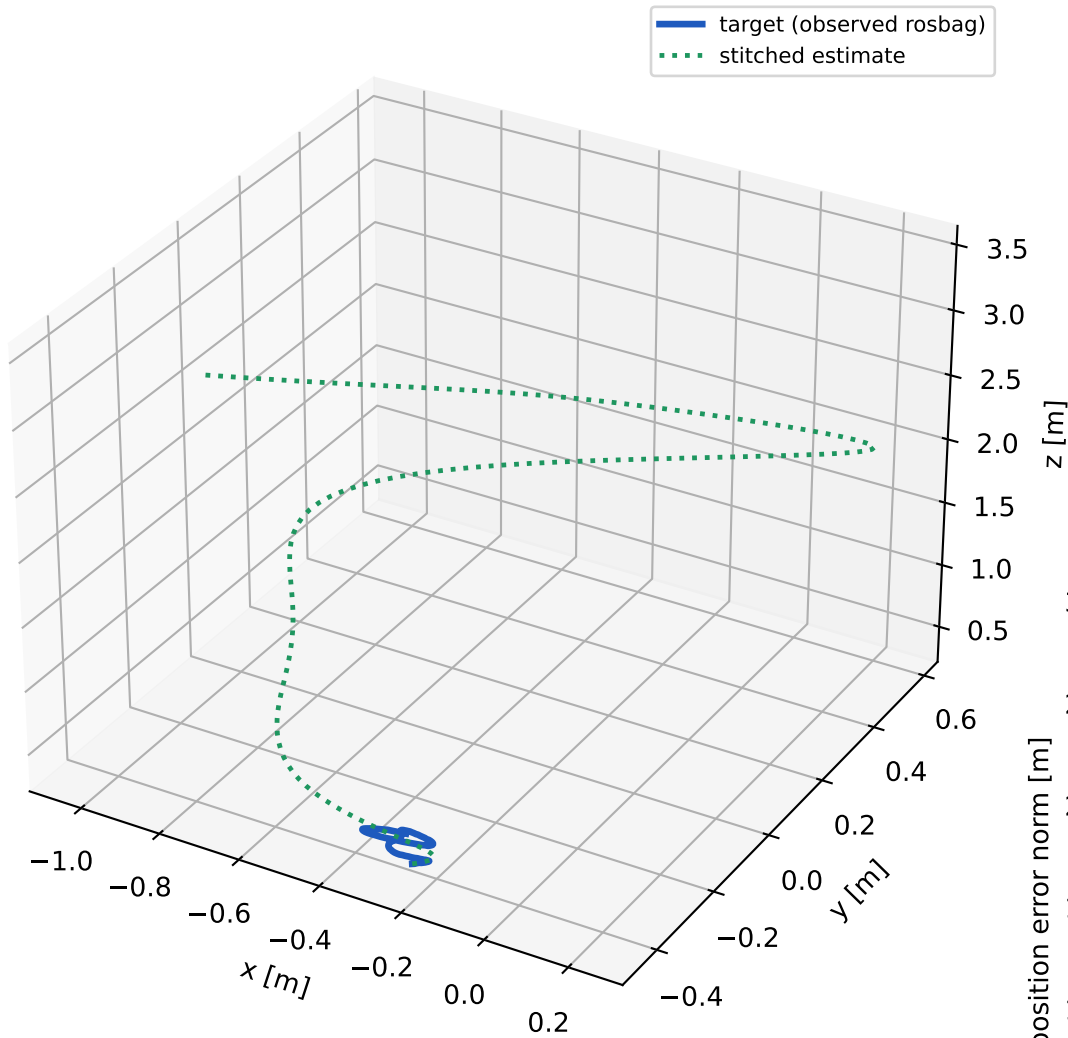
actual common support: 25.514–30.964 s  
selected command delay: 0.126554 s

| metric                                  | stitched |
|-----------------------------------------|----------|
| position RMSE [m]                       | 1.4105   |
| orientation RMSE [deg]                  | 13.087   |
| velocity RMSE [m/s]                     | 1.136    |
| angular velocity RMSE [rad/s]           | 0.42369  |
| specific force RMSE [m/s <sup>2</sup> ] | 0.65254  |

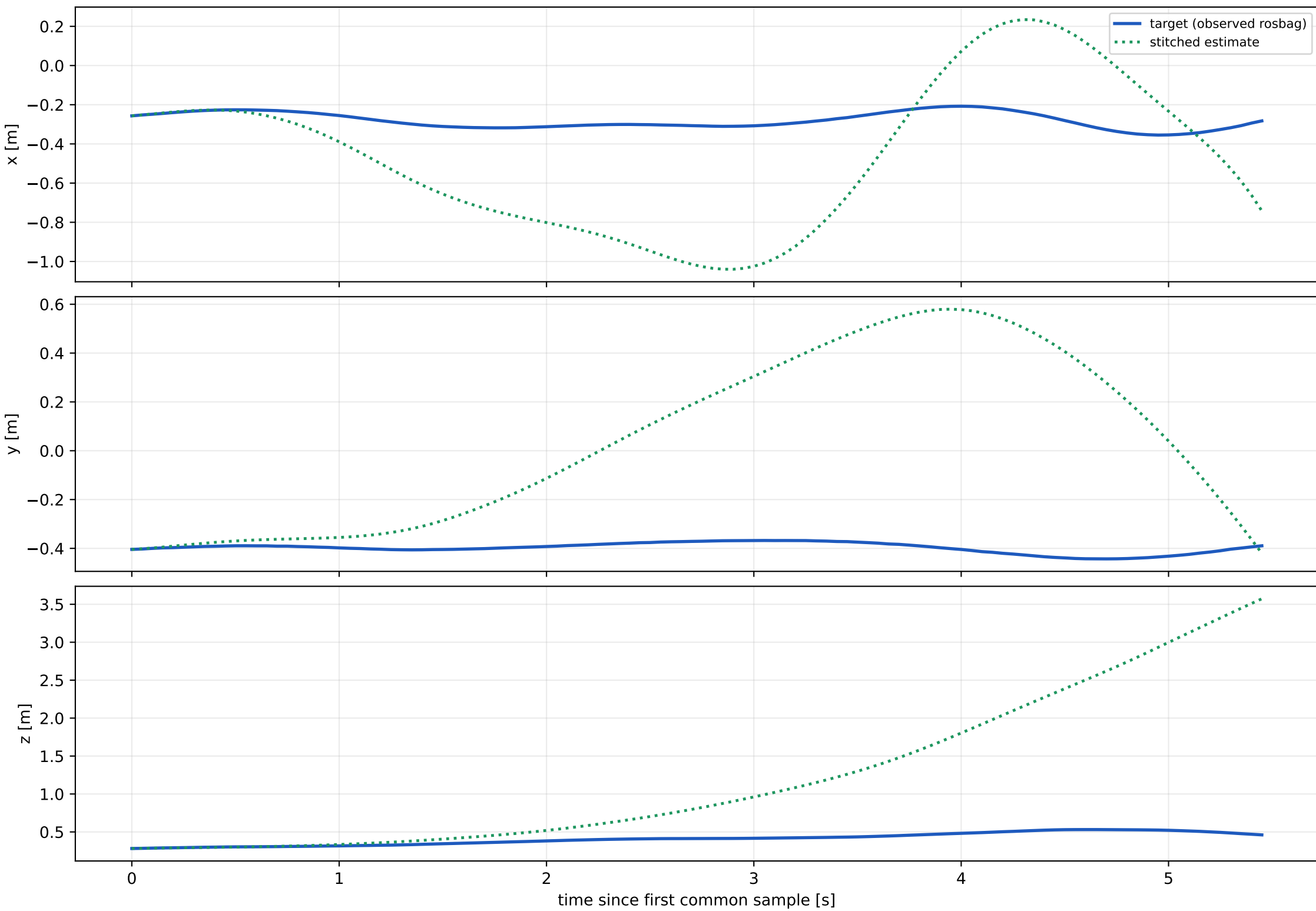
continuity max / tolerance:  
2.22e-15 / 0.001 (converged)

Velocity and IMU metrics are diagnostic only;  
the fit uses sensor position and orientation.

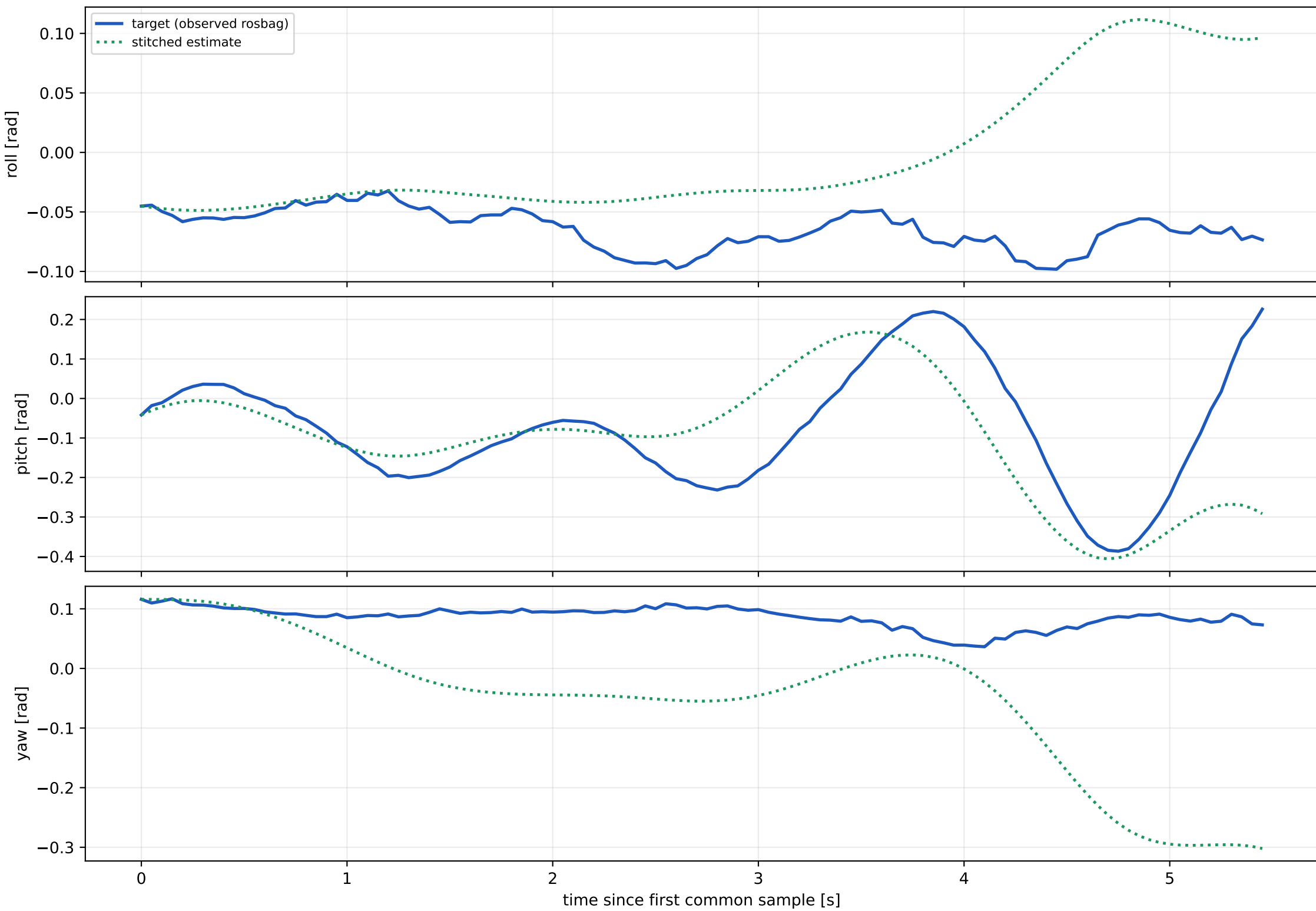
## Stitched trajectory



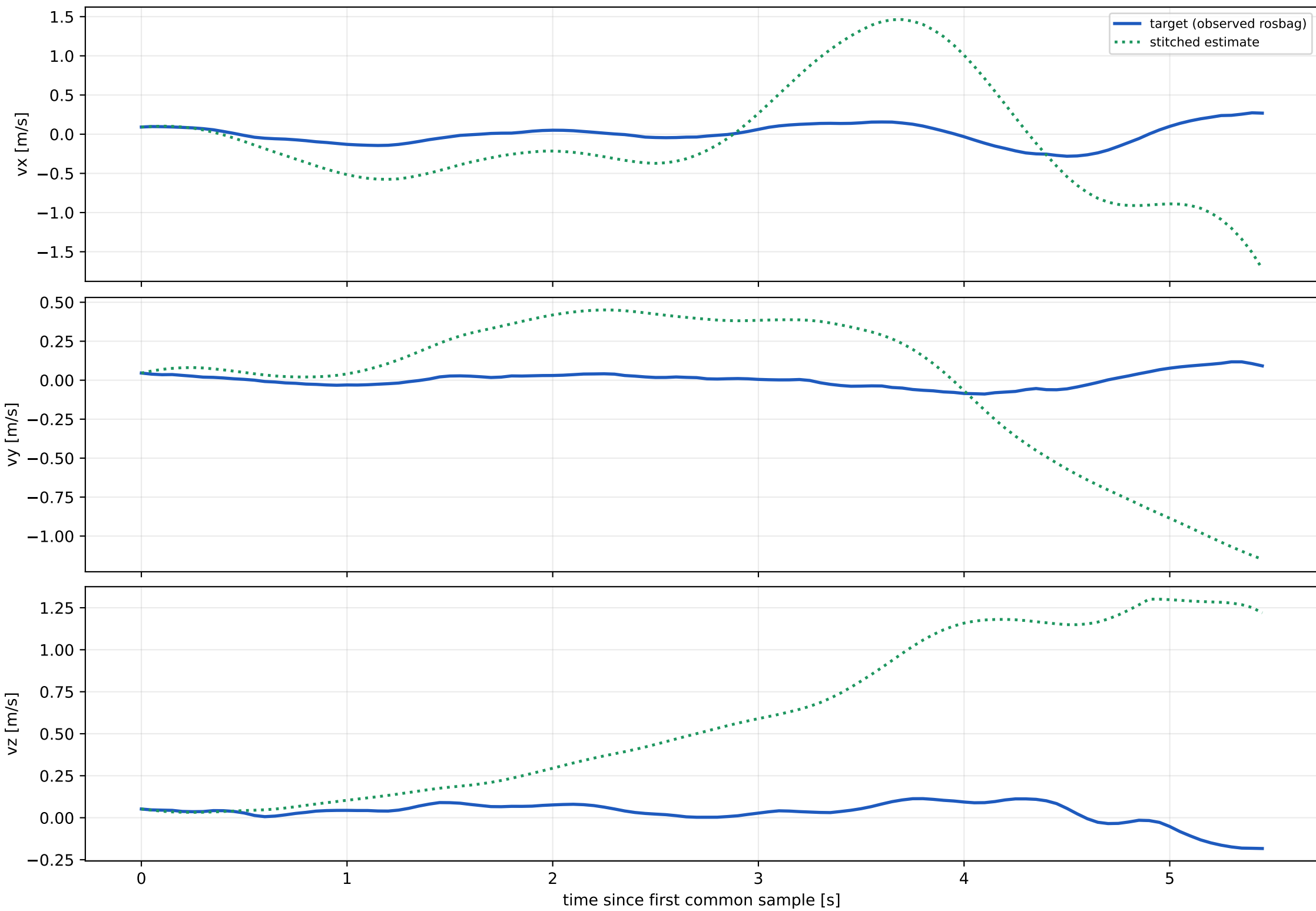
Sensor position



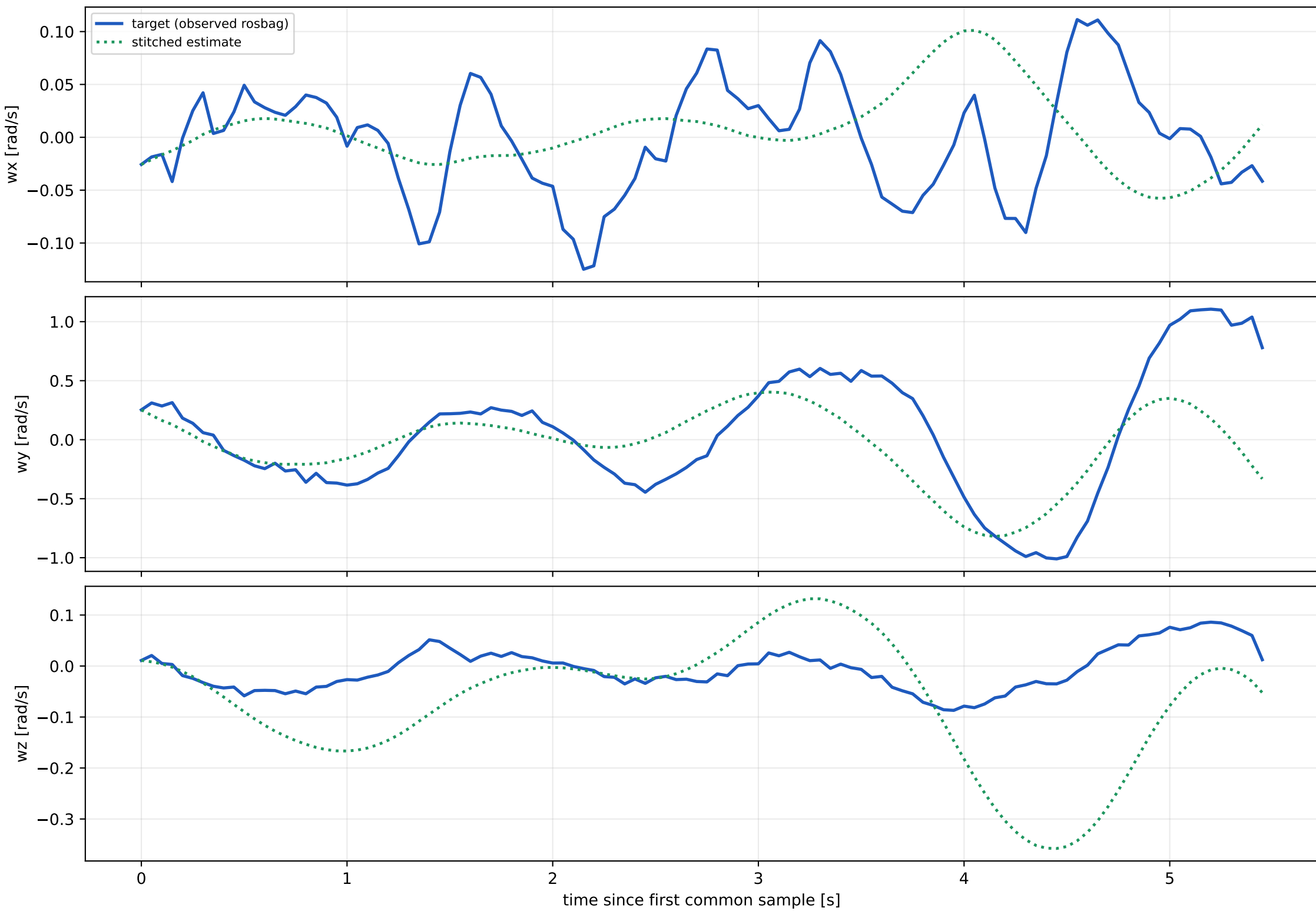
Sensor orientation (display only; fit uses SO(3) log)



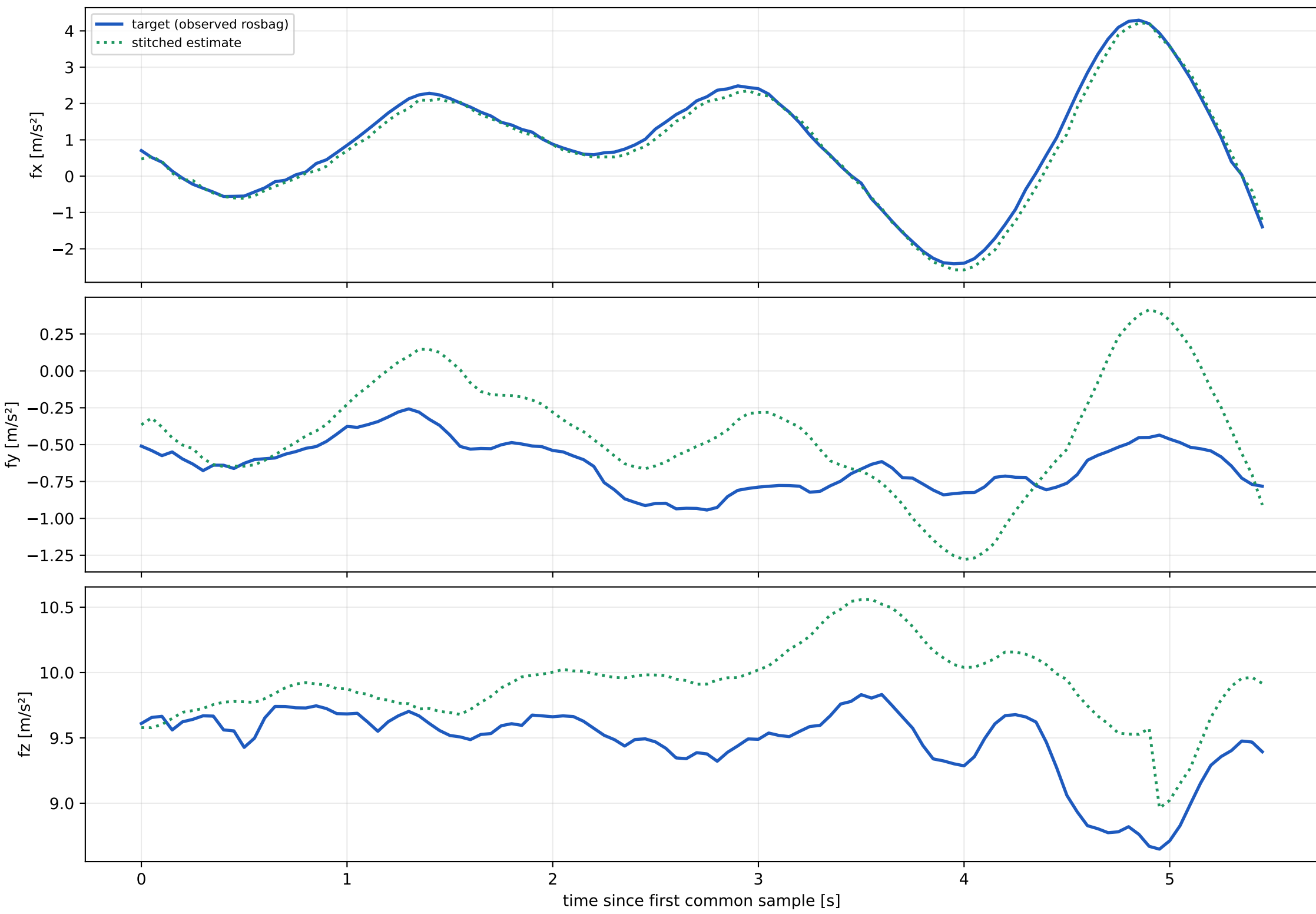
World-frame sensor velocity (diagnostic; excluded from fit)



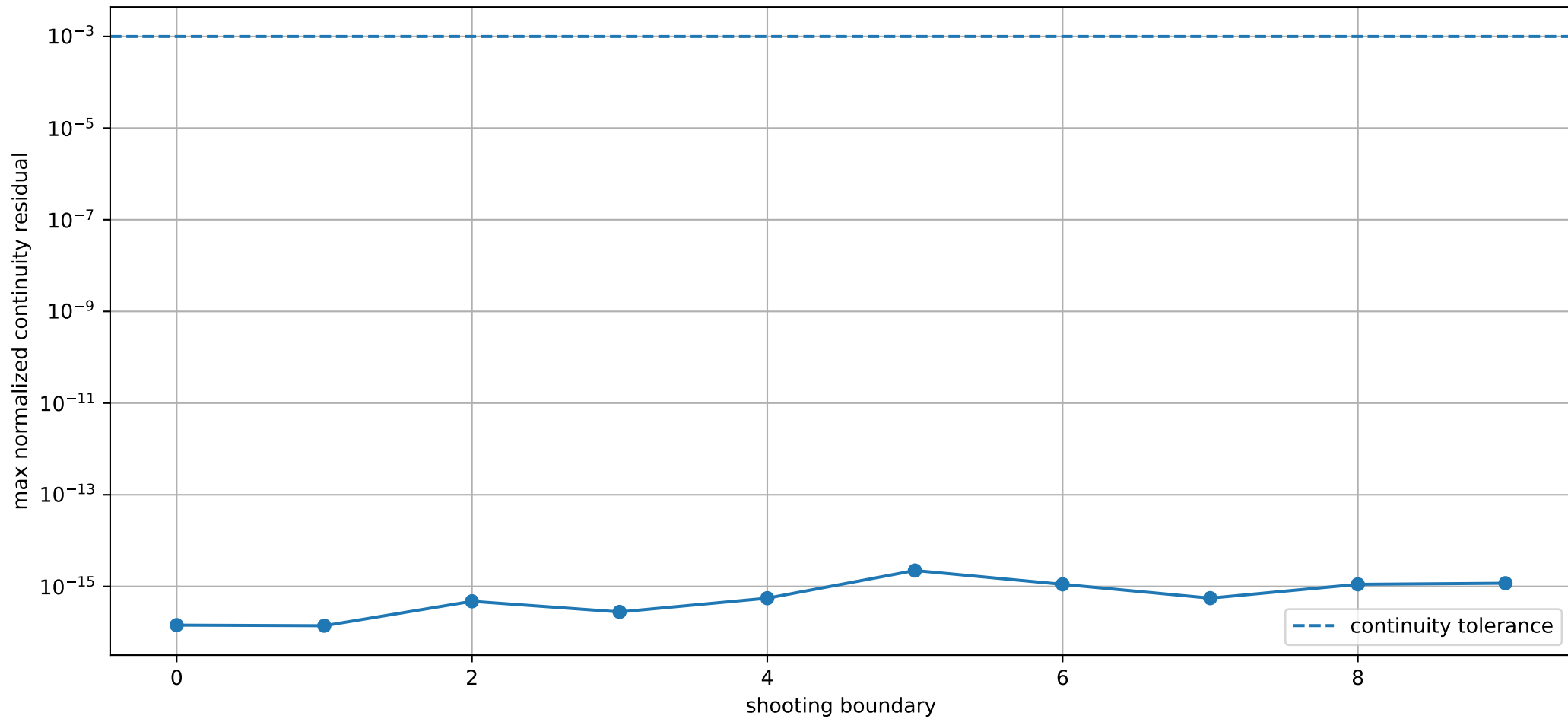
IMU angular velocity (diagnostic; excluded from fit)



IMU specific force (diagnostic; excluded from fit)



Shooting continuity diagnostic



# Final estimated parameters

## Selected multiple-shooting estimate

### Decoded physical parameters

```
mass [kg] 3.02232298411
inertia [kg m^2]
[ 0.969421478511 0.109415232744 0.00269418769725]
[ 0.109415232744 0.259121283672 -0.0238195866122]
[ 0.00269418769725 -0.0238195866122 1.22640781476]
principal inertia [kg m^2] [0.242065895329, 0.985890660947, 1.22699402067]
CoG offset [m] [0.0611824169108, -0.0936319079332, -0.0104546517358]
rotor force effectiveness [1.01939436606, 2.27207200437, 0.702843051126, 0.614295487471]
command delay [s] 0.126553995291
```

### Thirteen optimized smooth coordinates

```
log_mass_scale 0.250930812684
log_second_moment_cholesky_xx_scale 0.693465818191
log_second_moment_cholesky_yy_scale 1.32976612837
log_second_moment_cholesky_zz_scale 0.00131116299385
normalized_second_moment_cholesky_yx_offset -0.848128913489
normalized_second_moment_cholesky_zx_offset -0.070093160434
normalized_second_moment_cholesky_zy_offset 0.316546317295
cog_offset_x_m 0.0611824169108
cog_offset_y_m -0.0936319079332
cog_offset_z_m -0.0104546517358
force_effectiveness_contrast_1 -0.566734418026
force_effectiveness_contrast_2 0.630802486553
force_effectiveness_contrast_3 0.562661573802
```

### Fit diagnostics

```
stitched inertia-radius loss [m^2] 1.00161569946
full-rollout inertia-radius loss [m^2] 1.00161569946
broad soft-prior cost 0.947972727871
position RMSE [m] 1.41050708777
orientation RMSE [deg] 13.0867385912
velocity RMSE [m/s] 1.13600902673
angular velocity RMSE 0.423686515804 rad/s
specific force RMSE 0.652537512648 m/s^2
continuity max (normalized) 2.22044604925e-15
continuity tolerance 0.001
continuity converged True
```